AUTONOMOUS AI-DRIVEN HUMAN DETECTION IN HIGH-RISE BUILIDING FIRES USING YOLO

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Abstract:

High-rise building fires in densely populated urban areas claim numerous lives each year, and the absence of autonomous rescue systems forces firefighters to operate under extreme conditions of smoke, heat, and structural collapse. This project presents an advanced, AI-driven unmanned aerial vehicle (UAV) framework designed to optimize search-and-rescue operations in structural firefighting environments. Traditional visual search and reconnaissance methods frequently fail due to heavy smoke opacity and dense debris. By deploying YOLO_v8 or family-based object detection model directly on the system, real-time and ultra-low-latency inference is achieved. Furthermore, a live thermal video stream with overlaid computer vision bounding boxes and heat signature tracking provides incident command with immediate, actionable situational awareness. Ultimately, this integration of edge AI, localized thermal computer vision, and robust telemetry pipelines significantly reduces emergency response latency, minimizes human error, and enhances survivability in high-risk disaster zones. The proposed approach aims to eliminate the need for firefighter entry into hazardous zones, shorten rescue response times, and deliver a reproducible evaluation pipeline for autonomous aerial fire-rescue AI.

Keywords:

Computer Science and Software Engineering; Artificial Intelligence; Computer Vision; Human Detection in Smoke; High-Rise Building Fire Rescue; Autonomous UAV

Introduction:

The integration of artificial intelligence (AI), computer vision, and robotics is transforming fire rescue operations, particularly in hazardous high-rise building scenarios. Recent research demonstrates that AI-powered systems can autonomously identify and localize humans in dangerous environments, navigate complex structures, and minimize risks to human responders by leveraging advanced sensors (thermal and RGB), deep learning models (YOLO variants, CNNs), and real-time data processing. These solutions address challenges such as low visibility due to smoke, heat, and structural instability by fusing multimodal sensor data and employing robust classification algorithms. Ongoing research by well reputed scholars shows how different variants of the YOLO family models contribute to identifying humans in hazardous conditions. For instance, the "*Detection of Human Life in Fire Scenarios Using YOLOv8*" research paper, provides insights about YOLOv8 within the OpenCV framework: how the technology is used for image acquisition and video enhancement, and how images clearly feature humans enclosed in bounding boxes, isolating their position in the scene. This is a key finding for the project. The paper is a useful computational subsystem, but it lacks integration and on-board deployment evidence. This is crucial for the high-rise operational validation requirement of the project because the deployment phase is crucial. Notably, several studies have developed or referenced public datasets for training models in victim detection under fire conditions. This synthesizes the latest advancements in AI-driven autonomous rescue systems, in hazardous environments like a fire scene.

Methods:

A comprehensive literature search was conducted across over 170 million research papers indexed in Consensus, including Semantic Scholar and PubMed. The search strategy targeted foundational concepts, technical implementations of AI-driven rescue robotics, computer vision for human detection under fire conditions, adjacent domains (e.g., UAVs in disaster response), limitations/edge cases, and available datasets or benchmarks. In total, 7212 papers were identified; after relevance filtering and deduplication, 50 papers were included in this review.